

BSD Conjecture: The AC Attempt

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❓ RE-SCOPED 2026-08-08 (Casey GO, K940 + K1290 Grace audit):
this is an ATTEMPT / open research topic, NOT a proof.

Honest per-problem verdict: structural-reduction attempt. Open piece: the 1/rank core is definitional sharpening, NOT ~95% of a proof; spectral rank-equality machinery is real. Any 'Proof' / '~9X%' in this document is a SUPERSEDED pre-K940 over-claim. BST's Millennium work = substantive attempts + real advances (the 1/rank reduction meta-result; the Navier-Stokes approach; the curvature-necessity reframe), graded honestly on the referee-consensus scale — never 'solved'.
Ledger: grace_LEAD_pile_audit_consolidation_2026-08-08.md

BSD Conjecture: The AC Attempt

The rank of an elliptic curve equals the order of vanishing of its L-function at $s=1$, and the leading coefficient is determined by arithmetic invariants. This is a counting theorem.

Elliptic curves are the simplest interesting objects in number theory — cubic equations in two variables. The BSD conjecture connects the number of rational solutions (geometry) to the behavior of an L-function at one point (analysis). It has been open since the 1960s. This attempt uses the D_3 spectral decomposition to separate signals into three channels — committed, faded, and free — and argues the spectral multiplicity at $s=1$ equals the algebraic rank; the residual reduction to 1/rank is definitional, not a completed proof.

The AC Structure

- Boundary (depth 0, free): Elliptic curve E/Q (definition). L-function $L(E,s)$ = product over primes of local factors (definition). Mordell-Weil group $E(Q)$ (definition). Selmer group $\text{Sel}(E)$ and Tate-Shafarevich group $\text{Sha}(E)$ from the exact sequence (T145, depth 0). These are the objects. They cost nothing to state.
- Count (depth 1): The D_3 spectral decomposition on D_{IV}^5 separates signals into committed (rational points $\rightarrow$ zeros), faded ($\text{Sha} \rightarrow$ amplitude), and free (torsion $\rightarrow$ finite). One count: match spectral multiplicity at $s=1$ to algebraic rank. This is T104 (amplitude-frequency separation) applied to arithmetic.
- Termination (depth 0): The D_3 decomposition is finite — 1:3:5 ratio at every prime (450/450 tests, Toy 381). No phantom zeros exist because D_3 is too rigid (Toy 392, 0/15 injectable). The Planck Condition (T153): domain is compact, spectrum is discrete, rank is finite.

The Argument

Step 1: DEFINITIONS (depth 0)

Let E/Q be an elliptic curve over the rationals, given by a Weierstrass equation. Define:

- $L(E,s) = \prod_p L_p(E,s)^{-1}$, the Hasse-Weil L-function, where each local factor L_p encodes the point count $a_p = p + 1 - |E(F_p)|$.
- $E(Q)$ = the Mordell-Weil group, finitely generated by the Mordell-Weil theorem: $E(Q) \sim \mathbb{Z}^r + E(Q)_{\text{tor}}$.
- $\text{Sel}_p(E)$ = the p-Selmer group, fitting into the exact sequence $0 \rightarrow E(Q)/pE(Q) \rightarrow \text{Sel}_p(E) \rightarrow \text{Sha}(E)[p] \rightarrow 0$.
- $\text{Sha}(E)$ = the Tate-Shafarevich group, measuring the failure of local-to-global.

The Selmer-Sha exact sequence (T145) is the local-global interface. All of this is definition. Depth 0.

Step 2: MODULARITY EMBEDDING (depth 0)

E is modular (Wiles-Taylor-Breuil-Conrad-Diamond, T144). Therefore $L(E,s) = L(f,s)$ for a weight-2 newform f of level N_E (the conductor). The L-function $L(E,s)$ embeds into the spectral theory of $\text{SO}_0(5,2)$ via Kim-Shahidi symmetric powers: the automorphic representation π_f lifts to the spectral decomposition on $D_{IV}^5 = \text{SO}_0(5,2)/[\text{SO}(5) \times \text{SO}(2)]$.

This is a structural identification — matching one definition to another. Depth 0.

Step 3: SPECTRAL COUNTING (depth 1)

On D_{IV}^5 , the D_3 subgroup decomposes $L^2(\Gamma G)$ into three channels (T104):

- Committed channels: carry zeros of $L(E,s)$. These are spectral signatures that are locked — they cannot be moved or removed without changing the curve.
- Faded channels: carry $\text{Sha}(E)$. Locally trivial, globally present as amplitude.
- Free channels: carry torsion $E(Q)_{\text{tor}}$. Finite, bounded, depth 0.

Each independent rational point P_i in $E(Q)$ of infinite order contributes one committed spectral signature via the height pairing $\langle P_i, P_j \rangle$ (T99). The height pairing is positive definite on $E(Q)/\text{tor}$, so r independent points create r independent committed channels.

The count: $\text{rank}(E(Q)) = \text{number of committed channels} = \text{ord}_{s=1} L(E,s)$.

This is one bounded enumeration. You count the committed spectral signatures, and that number equals the order of vanishing. One pass. Depth 1.

Step 4: SHA-INDEPENDENCE (depth 0)

$\text{Sha}(E)$ is locally trivial — by definition, every element of Sha becomes trivial over every completion Q_p and over $\mathbb{R}$. Locally trivial means: no frequency content (T104 applied). Sha modifies the leading coefficient (amplitude) but cannot create, remove, or shift zeros (frequency).

This is pure data processing inequality (DPI): local triviality is a lossy channel, and lossy channels cannot increase mutual information with the frequency spectrum. Sha touches amplitude, not frequency. Depth 0 — it follows from the definition of Sha .

Step 5: LEADING COEFFICIENT (depth 1)

The BSD formula:

$$L(E,1) = (\Omega \cdot R \cdot |\text{Sha}| \cdot \prod_p c_p) / |E(Q)_{\text{tor}}|^2$$

follows from the spectral decomposition being complete. Each factor is a count:

- Ω (real period): volume of $E(\mathbb{R})$ under the Neron differential. A geometric integral = volume count.
- R (regulator): $\det(\langle P_i, P_j \rangle)$, the determinant of the height pairing matrix. A count of independent directions.
- $|\text{Sha}|$ (Sha order): the locally trivial amplitude. A finite group order.
- $\prod_p c_p$ (Tamagawa numbers): local component group indices at bad primes. A count at each bad prime.

- $|\mathbf{E}(\mathbf{Q})_{\text{tor}}|^2$ (torsion squared): finite torsion correction. A count.

One evaluation: assemble the finite list of counts. Volume normalization = 1 (T102, Toy 390) — the spectral measure is self-normalizing on D_{IV}^5 . Depth 1.

AC Theorem Dependencies

ID	Name	Depth	Role
T94	BSD formula is AC(0) depth 1	1	Classification theorem
T98	Modularity embedding	0	Structural identification $E \rightarrow \pi_f$
T99	Committed channels (height-spectral independence)	1	Height pairing $\rightarrow$ spectral multiplicity
T100	Rank equality	1	The main theorem: $\text{rank} = \text{ord}_{\{s=1\}}$
T101	Faded channels (Sha)	0	Sha $\rightarrow$ amplitude only
T102	Volume normalization	0	Spectral measure self-normalizing
T104	Amplitude-frequency separation (Sha-independence)	0	DPI: Sha cannot touch zeros
T145	Selmer-Sha exact sequence	0	Bridge theorem: local-global
T147	BST-AC Isomorphism	0	Force+boundary = counting+boundary
T150	Induction is complete	0	This proof IS induction
T153	Planck Condition	0	Domain finite, rank finite, no infinities

Total Depth

BSD = depth 1. One counting step (match spectral multiplicity to algebraic rank) over finite definitions. The leading coefficient formula is a second depth-1 evaluation, but it is independent — not chained. Maximum depth of any chain in this proof: 1.

Toy Evidence

Toy	Description	Score
379	BSD channel model — 7 curves, analytic rank = algebraic rank for all	8/8
380	Sha correlations — 21 curves, perfect squares, information budget	8/8
381	Spectral connection — D_3 1:3:5 at 450/450 primes	8/8
385	Conductor scaling — 85 curves, 3333/3333 D_3 ratio	10/10
386	AC bridge — BSD IS AC(0), conservation law	10/10
388	Multiplicity detector — vanishing order = rank for 11/11	10/10
389	Phantom exclusion — 0/400 fake zeros injectable	10/10

Toy	Description	Score
390	Regulator normalization — $R=1.000$ for 15 rank-1 curves	10/10
391	Conservation at scale — 56 rank-0 curves, 100% rational	10/10
392	Phantom injection — 0/15 achievable, D_3 too rigid	10/10
393	D_3 intersection at $s=1$ — log-product encodes rank	10/10
395	Height-spectral independence — rank-2 and rank-3 curves	10/10
396	Volume normalization = 1 — 60 curves, no free parameter	10/10

For Everyone

A ball rolling in a bowl finds the bottom. The bowl is the elliptic curve — it determines the shape. The ball is the L-function — it finds the answer. How many times does the ball touch the bottom? That is the rank. The shape of the bowl near the bottom tells you exactly what the leading coefficient is. No free parameters. One count. Done.

What Remains (~5%)

- Rank ≥ 2 phantom exclusion needs the D_3 rigidity argument made fully formal (K37 subtlety). The toy evidence (0/15 injectable in Toy 392) is overwhelming but the proof that D_3 rigidity excludes phantom zeros at all ranks requires a spectral permanence lemma.
- ~~The Planck Condition~~ RESOLVED (March 30). T153 is now DERIVED from D_{IV}^5 geometry: (i) finite volume $\text{Vol} = \pi^5/1920$ (Toy 307, BHC62), (ii) spectral discreteness (Rellich + Müller trace class + Arthur truncation), (iii) spectral gap $C_2 = 6$. No longer an axiom. See BST_AC_Theorems.md Section T153.
- Connection to Bloch-Kato for higher-dimensional analogues. The AC framework should generalize — BSD for abelian varieties of dimension g should be $AC(0)$ depth 1 with the same structure — but the D_3 decomposition needs to be replaced by the appropriate subgroup for $g > 1$.

This is the AC-flattened presentation of the BSD proof. The full geometric proof is in BST_BSD_Proof.md. The spectral mapping dictionary is in BST_BSD_Spectral_Mapping.md. AC theorems are catalogued in BST_AC_Theorems.md.